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Influence of high-temperature CO₂ in hot flue gas on the wettability of coal surface: Experimental and molecular simulation study

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Highlights

- The influence of competitive adsorption of multiple gases on wettability is discussed.
- Comparison of contact angle with other wettability parameters.
- The stage variation of coal wettability with gas injection depth is studied.

Abstract

Competitive wetting is a key factor influencing capillary forces, CO₂, and water adsorption, and gas desorption and migration. Aiming at uncovering the patterns and mechanisms of water wettability changes with coal pores and fractures under high-temperature and high-pressure CO₂ injection (333–413K, 4MPa), this paper presents experiments on the T₂ spectra of moisture within coal pores and fractures, the adsorption characteristics of CO₂ in coal, and the coal-water-gas interfacial property parameters, an analysis on Scanning Electron Microscope (SEM) images of coal surfaces treated with CO₂ at different temperatures, as well as molecular dynamics simulations on CO₂ injection into slits of water-bearing rough coal. The experimental and simulation results disclose that, under high-temperature CO₂ injection, the water wettability within coal pores changes in three stages from the injection well toward the interior of the coal seam, characterized by a “decrease–increase–decrease” pattern. These three stages are governed by different mechanisms: initially by strong activation arising from the temperature field, followed by moderate activation, and finally by CO₂-moisture competitive wetting. Additionally, a water film of appropriate thickness on the coal surface is beneficial for CO₂ adsorption. The three-phase contact angle (θ) cannot serve as a sole basis for accurately judging changes in water wettability within the flow channels of micropores and fractures in coal. Instead, it only effectively reflects changes in droplet wettability caused by competitive adsorption of CO₂ molecules on the coal surface. This study provides guidance on the optimal injection temperature for hot flue gas, which is beneficial for CO₂ capture and improves the extraction efficiency of CBM.

Keywords

Wettability stage changes; Multi-phase interface; Molecular dynamics; High temperature adsorption; CO₂ capture

CRediT authorship contribution statement

Declaration of competing interest

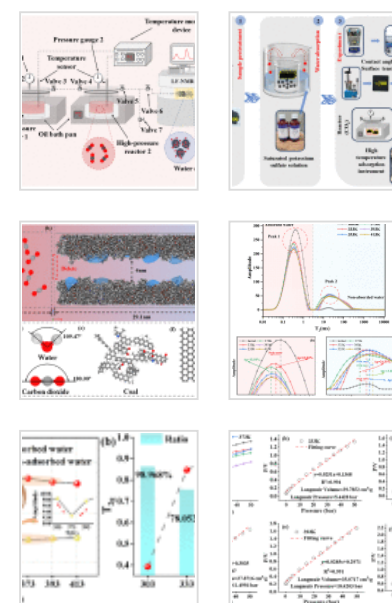
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1. Introduction

Table 3. Surface tension data.

Coalbed methane (CBM), one of the major mineral resources in China, boasts substantial reserves in deep formations, yet its extraction poses significant challenges. The primary technical bottleneck lies in the low permeability of coal seams and the difficulty in methane desorption and flow, which consequently lead to low extraction efficiency (Espinoza et al., 2015; Guanhua et al., 2019). In recent years, the use of hot flue gas from coal-fired power plants to enhance CBM extraction has garnered considerable attention in the industry. The hot flue gas primarily comprises high-temperature and high-pressure N₂ and CO₂.

Competitive wetting between high-temperature and high-pressure CO₂ in hot flue gas and inherent moisture in coal seams is a crucial factor controlling capillary forces, CO₂ and water adsorption, and methane desorption and migration. It directly influences the effectiveness of hot flue gas in enhancing the permeability of low-permeability coal seams and displacing CBM, as well as the geological sequestration capacity of CO₂ (Cai et al., 2024; Lei et al., 2022). Mahoney et al. measured the contact angle of the H₂O-CO₂-anthracite system under a pressure of 0.00–140.00bar, and found that the anthracite surface is water-wet at atmospheric pressure, but it transitions to CO₂-wet when the pressure exceeds 26.00bar (Mahoney et al., 2017). Arif et al. measured the CO₂ wettability of different-rank (from low-rank to high-rank) coal samples by the pendant drop method, and reached the following conclusions: High-rank coal exhibits strong CO₂ wettability; medium-rank coal shows weak CO₂ wettability; and low-rank coal displays intermediate wettability under typical reservoir conditions. Besides, the CO₂ wettability of coal strengthens with the increase of pressure and salinity but decreases with the rise of temperature (Arif et al., 2016). Low-field nuclear magnetic resonance (LF-NMR) relaxation time analysis can measure the relaxation time of hydrogen-containing fluids (¹H) in the micropores of coal samples, enabling the study on wettability patterns in such micropores (Cai et al., 2023; Mao et al., 2020; Zheng et al., 2020b). Yao et al. investigated the mechanisms by which pressure, temperature, and water state influence CO₂-H₂O wettability during CO₂-enhanced coalbed methane (ECBM) recovery (Zheng et al., 2020a).

Through a defined method for analyzing bound water, Zhu et al. discovered that CO₂ pressure hinders the conversion of free water to adsorbed and capillary water, and the wetting equilibrium time shortens with the rise of CO₂ pressure (Zhu et al., 2024). Sun et al. estimated the contact angle by changes in the T_{2g} value of the free water peak, so as to analyze the CO₂ and water wettability of coal (Sun et al., 2018). Molecular dynamics simulation, as a research method at the micro- and nanoscale, has been widely applied by scholars in the field of wettability (Ma et al., 2024a, 2024b; Zhao et al., 2024). Ma et al. explored the dynamic changes in wettability within nanoscale slits in coal during hot flue gas injection, and concluded that appropriately raising the injection pressure of hot flue gas is conducive to the shift from water-wet to gas-wet and the relief of the water-locking effect (Ma et al., 2024c). By constructing a CO₂-H₂O-kerogen system, Zhou et al. yielded the finding that the CO₂ film on the kerogen surface hinders the spreading of water molecules (Zhou et al., 2022).

Existing researches are primarily focused on the wettability changes of modified coal, yet the modification treatments do not align with real injection processes. Moreover, the coupling between the thermal effects induced by the injection of high-temperature and high-pressure CO₂ into water-bearing coal and the competitive wetting effects of CO₂ has not been clarified yet, especially regarding how the two factors influence coal wettability as the injection depth increases. In this study, the variations in wettability within coal pores and fractures under CO₂ injection at different temperatures were explored from both macroscopic and microscopic perspectives through a combination of experimental and molecular dynamics methods. Furthermore, the underlying mechanisms driving the three-stage changes in water wettability with the increase of injection depth were revealed.

2. Experimental testing and molecular dynamics simulation

2.1. Coal sample preparation and analysis

The coal samples used in this experiment were sourced from Wulanmulun (WLML) Coal Mine in Inner Mongolia, China. The lump coal was crushed using a coal breaker into two particle sizes. In LF-NMR experiments, coal particles (60–80 meshes) were employed as experimental samples for the purpose of achieving a balance between water absorption efficiency and measurement precision (Wang et al., 2020; Zheng et al., 2020a). In contact angle measurements, small coal particles (>200 meshes) were adopted for pressing coal pellets for the purpose of minimizing the effect of surface heterogeneity and obtaining smoother surfaces for stable and accurate measurements at different temperatures (Sun et al., 2018; Wang et al., 2023). The 4mm×4mm×2mm coal blocks were used for SEM analysis. The experimental coal samples were subjected to proximate analysis in accordance with the “Method for Proximate Analysis of Coal” (GB/T 212–2008). Meanwhile, the basic physical properties of the coal samples were measured following the “Method for Microscopical Determination of Vitrinite Reflectance of Coal” (GB/T 6948–2008). The test results are presented in Table 1.

Table 1. Results of proximate analysis on the coal sample.

Coal sample	Proximate analysis				Maximum vitrinite reflectance/%
	M _{ad} /%	A _d /%	V _{daf} /%	FC _{ad} /%	
WLML	0.53	9.58	11.19	80.31	2.81

2.2. LF-NMR method and experiment

2.2.1. LF-NMR method

LF-NMR is a non-destructive and continuous detection method whose transverse relaxation time T_2 is often used to characterize fluid information in porous materials. The T_2 distribution is governed by three relaxation mechanisms, namely free relaxation, surface relaxation, and diffusion relaxation. For coal samples, the effects of free relaxation

and diffusion relaxation on T_2 can be neglected (Yao and Liu, 2012), then (Zhang et al., 2022)

$$\frac{1}{T_2} \approx \rho_2 \frac{S}{V} = F_s \frac{\rho_2}{r} \quad (1)$$

where T_2 is the surface relaxation time; ρ_2 is the surface relaxation rate; S is the surface area of pores; V is the volume of pores; and F_s is the geometric factor. For spherical, cylindrical, and slit-shaped pores, F_s equals 3, 2, and 1, respectively, where $C=F_s\rho_2$. Then, Eq. (1) can be transformed into:

$$r = F_s \rho_2 T_2 = C T_2 \quad (2)$$

For the same type of coal, C is a fixed constant, and r is positively correlated with T_2 , that is, a longer T_2 corresponds to a larger r value. The occurrence characteristics of fluids in porous media are commonly quantitatively characterized by changes in the weighted average value of T_2 , denoted as T_{2g} . A smaller T_{2g} indicates that the fluid is present in smaller pores (Zhang et al., 2021). T_{2g} is obtained using Eq. (3) (Liu et al., 2020):

$$T_{2g} = \exp \left(\sum \frac{\ln(T_{2i}) A_i}{A_{Total}} \right) \quad (3)$$

where T_{2i} is the transverse relaxation time at point i ; A_i is the signal amplitude of T_2 at point i ; and A_{Total} is the total amplitude of the T_2 spectrum.

2.2.2. LF-NMR experiment

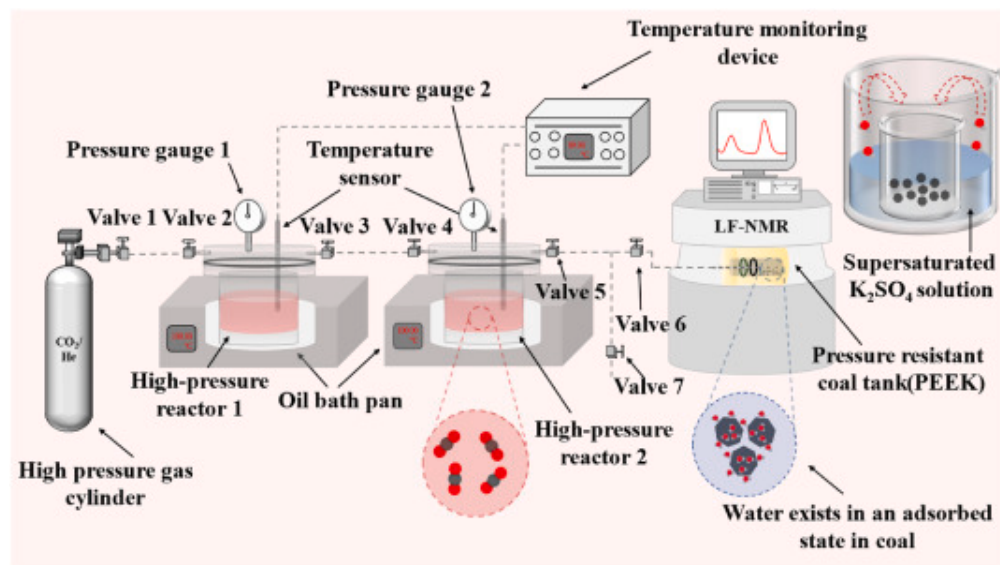
As shown in Fig. 1, the entire experimental system consists of an LF-NMR testing instrument, a PEEK container, a high-pressure reactor, a temperature controller, an oil bath, and a CO₂ cylinder. The relaxation information of fluids in the coal samples was analyzed by the MesoMR23-060H NMR analyzer (Newmark Analytical Instruments, China). Some parameters of the NMR analyzer are provided here: a magnetic field intensity of 0.50–0.08T, a main frequency of 21.30MHz, a magnet temperature of 32.00±0.02°C, and an echo time of 0.10 ms. Before CO₂ injection, the airtightness of the

entire system was checked thoroughly. A preconditioning reactor (reactor 1) and its internal temperature sensor were employed during the experiments for ensuring that the injected CO₂ reached the target temperature and pressure and for maintaining a stable internal pressure (4MPa) in the PEEK container after injection. The specific experimental procedure is introduced as follows.

- a) First, 10.00g of dried coal powder with a particle size of 60.00–80.00 meshes was placed into a small beaker, and the entire assembly was weighed. Subsequently, the small beaker was placed inside a larger beaker that had been filled with 200.00ml of supersaturated K₂SO₄ solution. The larger beaker was then tightly sealed with plastic wrap. The entire assembly of the small beaker was weighed every 12.00h until the change in mass was less than 0.01 g.
- b) Valves 1, 2, 3, 4, 5, and 6 were opened. Once pressure gauges 1 and 2 stabilized, valve 1 was closed. The airtightness of the system was monitored by observing the values on pressure gauges 1 and 2 every 60.00min. All valves were opened except valve 1, and all valves were closed once the values on pressure gauges 1 and 2 dropped to 0.
- c) The wetted coal powder was transferred into the PEEK container, and its T₂ spectrum was measured every 120.00min until the amplitude no longer changed.
- d) The oil bath temperature was set to 333.00K, and valves 1 and 2 were opened. Once the temperature monitored by the temperature sensor was the same as that of the oil bath, valves 1 and 2 were closed, and valves 3 and 4 were slowly opened. When the pressure on gauge 2.00 rose to 4.00MPa, valves 3 and 4 were immediately closed. Subsequently, valves 5 and 6 were opened, and the pressure on gauge 2 was maintained at 4.00MPa by finely adjusting valve 3. The T₂ spectrum was measured every 120.00min until the

amplitude no longer changed. Valves 3 and 7 were then opened to reduce the pressure gauge readings to 0.00. Finally, all valves were closed.

- e) Step d) was repeated with the oil bath temperature set to 353.00K, 373.00K, 393.00K, and 413.00K, respectively.



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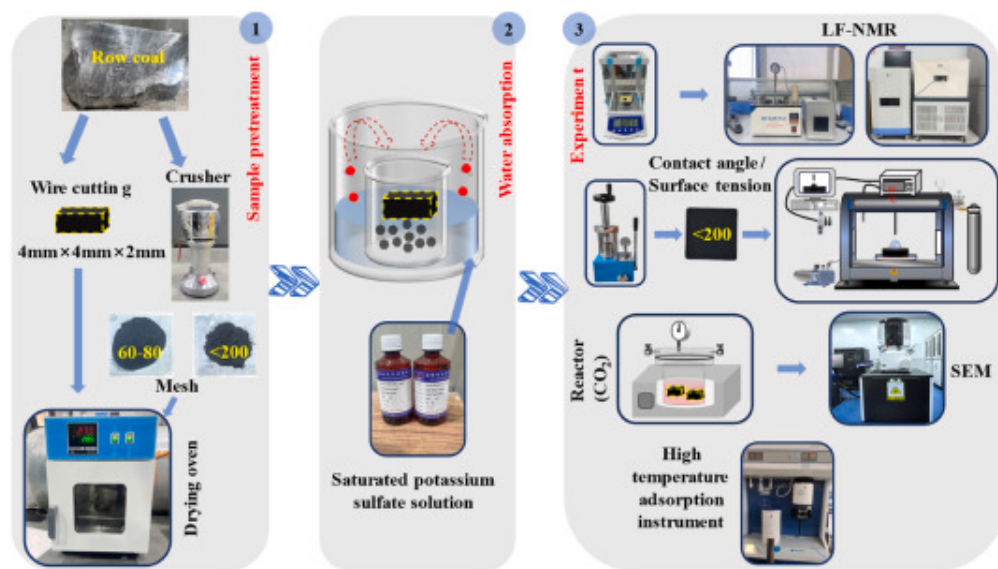
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Fig. 1. LF-NMR experimental system.

2.3. High-temperature and high-pressure contact angle and surface tension experiment

In this study, the coal-water-gas contact angle and the surface tension of droplets in a CO₂ atmosphere were determined by a DSAHT17C high-temperature and high-pressure contact angle measuring instrument (KRSS, Germany) (Fig. 2). In the heating system, silicon molybdenum rods were employed as the heating material, and the temperature

was raised through programmed heating at a rate of 20.00°C/min. The accuracy of contact angle measurement was 0.01°. The optical system was equipped with a 6-times zoom lens and a high-intensity LED light source. The observation lens, made of sapphire, was resistant to contamination and could withstand high temperatures and high pressures. The test pressure in this study was set to 4.00MPa, and the test temperature ranged from 333.00K to 413.00K, with a gradient of 20.00K. In both surface tension and contact angle experiments, three measurements were made at each temperature, and the average of the three measurements was taken as the final result.



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Fig. 2. Experimental procedure.

2.4. High-temperature and high-pressure CO₂ adsorption experiment

The adsorption behavior of CO₂ within coal at pressures of 0.00–5.00MPa and temperatures of 333.00–413.00K was determined with the aid of a fully automated high-temperature and high-pressure gas adsorption analyzer (BSD-PHD, Beishide, China) ([Fig.](#)

2). The adsorption was considered to have reached an equilibrium state when the pressure change remained below 0.025 bar within a continuous period of 1800.00s, the maximum adsorption equilibrium duration being 3600.00s for each data point. The degassing duration was 1200.00min, and the degassing temperature corresponded to the adsorption temperature. The temperature range was measured using He, and the volume of the temperature range was also tested.

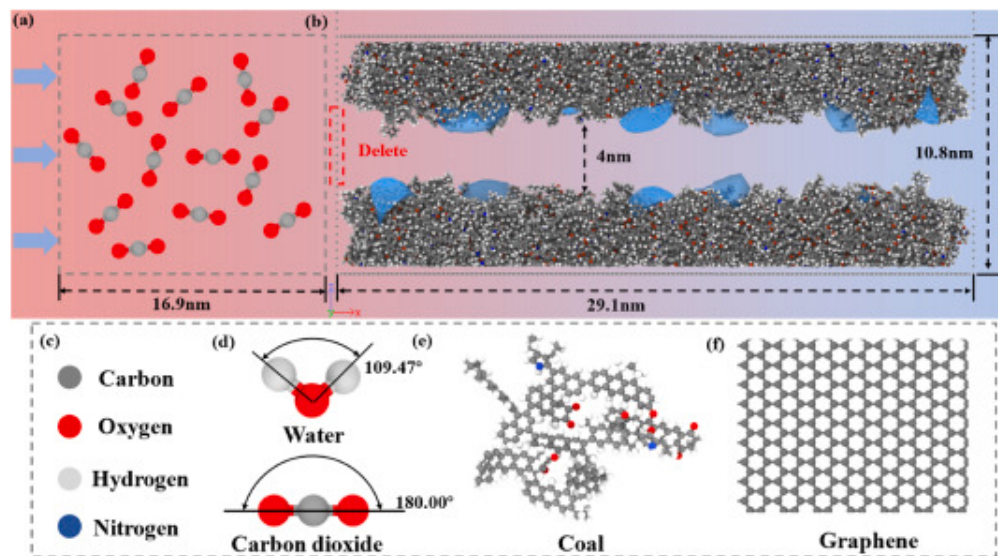
2.5. Molecular dynamics simulation

2.5.1. Force field theory

The DREDING force field ([Mayo et al., 1990](#); [Nosé, 1984](#)), a commonly adopted general force field capable of accurately describing complex intermolecular interactions, was employed for constructing the macromolecular model of coal ([Li et al., 2021](#); [Wu et al., 2019](#)). The structure and composition of these macromolecules were validated based on the results of ultimate analysis, Fourier transform infrared spectroscopy, and nuclear magnetic resonance carbon spectroscopy on real samples ([Feng et al., 2021](#); [Ping et al., 2020](#); [Wang, 2021](#)), with a molecular formula of C₁₉₉H₁₄₆N₂O₉. The SPC/E force field ([Berendsen et al., 1987](#)), known for its excellent performance in representing water's structural ([Yang et al., 2022](#)), spatial, and dynamic properties, and interactions with other molecules over a wide range of temperatures and pressures, was employed to describe the behavior of water molecules in this study. In addition, carbon dioxide molecules were described by the EPM_2 force field which accounted for polarization effects ([Zhu et al., 2009](#)), so as to provide more accurate intermolecular interactions under high-temperature and high-pressure conditions ([Aimoli et al., 2014](#); [Stubbs, 2016](#)).

2.5.2. Model construction

The model used in this study is illustrated in [Fig. 3](#). In the light of a modular modeling approach, the entire dynamic wettability change model consisted of three components, i.e., high-temperature and high-pressure CO₂, a slit model, and substances within the slit.



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Fig. 3. Molecular model for depicting dynamic wettability changes.

The same method as in previous study was adopted to ensure that the internal stress of the high-temperature and high-pressure CO₂ model matched the external pressure. Meanwhile, CO₂ was separately relaxed in the NVT ensemble for 2 ns to minimize the system energy.

The rough media slit was realized by adopting an NPT ensemble and an NVT ensemble with unilateral graphene plate constraints. The relaxation time was 2 ns under both ensembles. After relaxation, the coal density fluctuated slightly around 1.47 g/cm³, which lay within the range of real anthracite density. The relaxed single-layer rough coal was then duplicated, moved upward, and rotated 180.00° to construct the rough slit model. The dimensions of the rough slit model were 291.00Å×50.00Å×108.00Å, and the slit was 40Å wide on average.

The separately relaxed moisture-containing slit model was combined with CO₂ models at different temperatures. To ensure the interconnection between CO₂ and the slit model and to guarantee the smoothness of the injection process, the carbon atoms at a height of 35.00–75.00 Å were removed before the injection based on the spatial positions of the upper and lower coal seams in the Z-direction at the left entrance of the slit model. In this way, the construction of the overall model was completed. The model dimensions were 460.00 Å × 50.00 Å × 108.00 Å.

2.5.3. Molecular dynamics

The entire simulation was conducted in LAMMPS ([Plimpton, 1995](#)). After combination of the model, the NVE ensemble was applied to the entire simulation box, which was then run for 5.00 ns with a time step of 1.00 fs. During the simulation, the Shake algorithm was used to maintain the rigidity of water molecules ([Kräutler et al., 2001](#)). The Lennard-Jones (12-6 LJ) potential and the short-range Coulomb force with a cutoff radius of 12.00 Å were employed to describe non-bonded interactions between atoms:

$$U(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q^j q^i}{4\pi\epsilon_0 r_{ij}} \quad (4)$$

where r is the distance between two atoms; ϵ is the depth of the potential well; and σ is the distance between atoms when the LJ potential equals 0 ([Jones and Chapman, 1924](#)). The Particle-Particle-Particle-Mesh (PPPM) solver was used to calculate the long-range Coulomb force, with a computational accuracy of 1×10^{-4} ([Brown et al., 2011](#)). The Lorentz-Berthelot mixing rule was adopted to calculate non-bonded interactions between different types of atoms:

$$\sigma_{\alpha\beta} = (\sigma_{\alpha} + \sigma_{\beta}) / 2 \quad (5)$$

$$\epsilon_{\alpha\beta} = \sqrt{\epsilon_{\alpha}\epsilon_{\beta}} \quad (6)$$

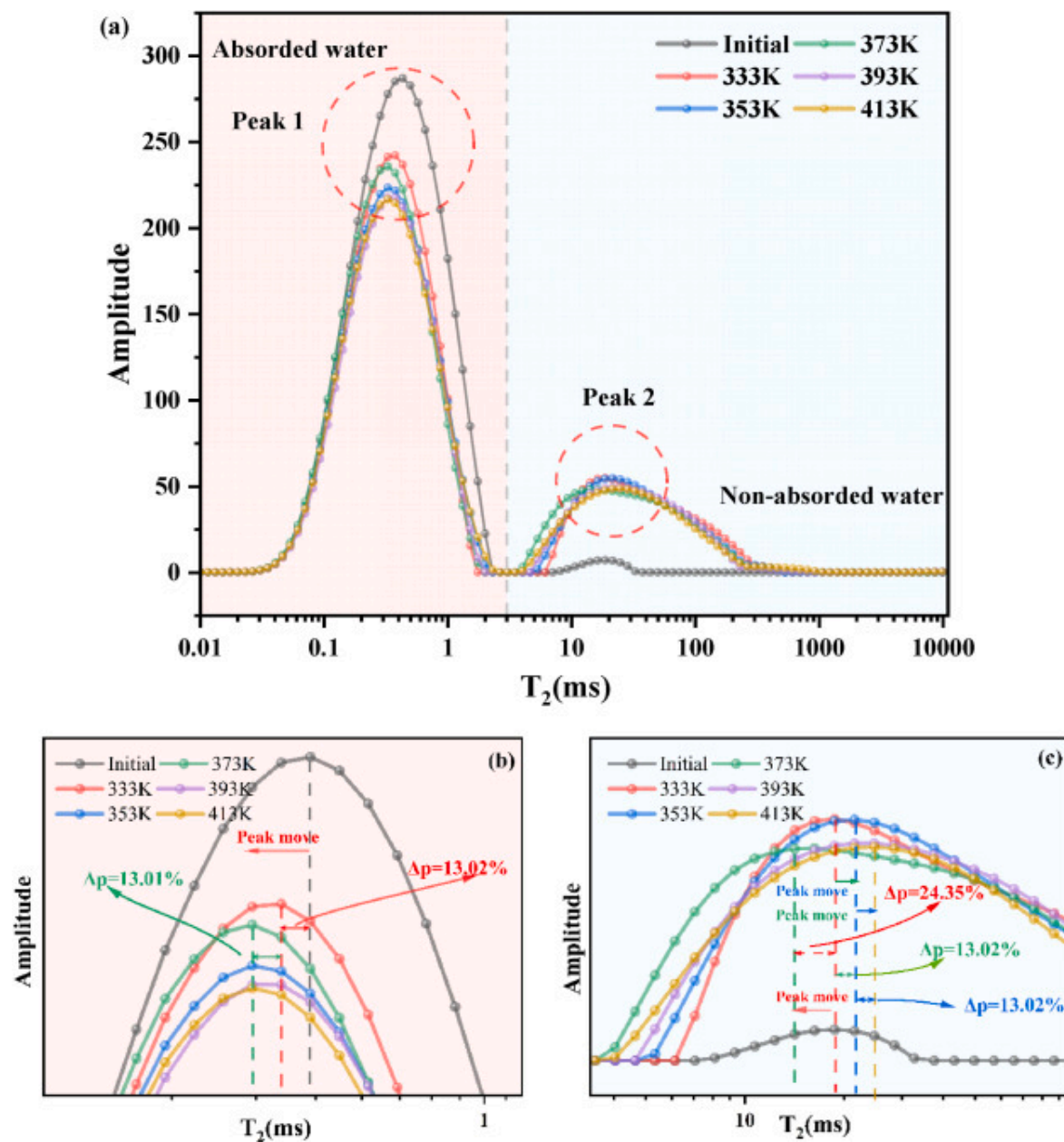
where α and β are two different types of atoms. Periodic boundary condition (PPP) was applied to all model boundaries, and the coal body remained fixed during the simulation.

3. Results and discussion

3.1. Analysis on experimental results

3.1.1. LF-NMR analysis

The relaxation time of water in coal varies depending on its occurrence state. Specifically, the relaxation time T_2 is shorter than 10.00 ms for adsorbed water, between 10.00 ms and 100.00 ms for capillary water, and longer than 100.00 ms for free water. Capillary water and free water are collectively referred to as non-adsorbed water. To enhance the visibility of changes in the occurrence state of water after high-temperature CO₂ injection, a humid environment was created using an oversaturated K₂SO₄ solution, which encourages more water to be adsorbed onto the coal surface rather than exist as free liquid water, as it would under immersion. Fig. 4 gives the stable T_2 spectra before CO₂ injection and after CO₂ injection at different temperatures. An evident peak for adsorbed water appears on the left side of $T_2=3.00$ ms, while another peak for non-adsorbed water appears on the right side of it. As can be seen in Fig. 4(b) and (c), compared to the case before CO₂ injection, when CO₂ is injected at 333.00K, the adsorbed water peak falls significantly, while the non-adsorbed water peak rises notably. Additionally, the position of the adsorbed water peak amplitude shifts leftward by 13.02%. As the injection temperature gradually rises, the adsorbed water peak exhibits a trend of first decreasing, then increasing, and finally decreasing again, the increase being quite anomalous at 373.00K. After the position of the adsorbed water peak amplitude continues to shift leftward by 13.01% at 353.00K, it no longer changes obviously. The non-adsorbed water peak does not change significantly with the rise of CO₂ injection temperature, but the position of the non-adsorbed peak amplitude is more sensitive to temperature changes. Compared to 333.00K, the position of the non-adsorbed peak amplitude shifts rightward by 13.02%, 24.35%, and 24.36% at 353.00K, 393.00K, and 413.00K, respectively, while it shifts leftward by 24.35% at 373.00K.

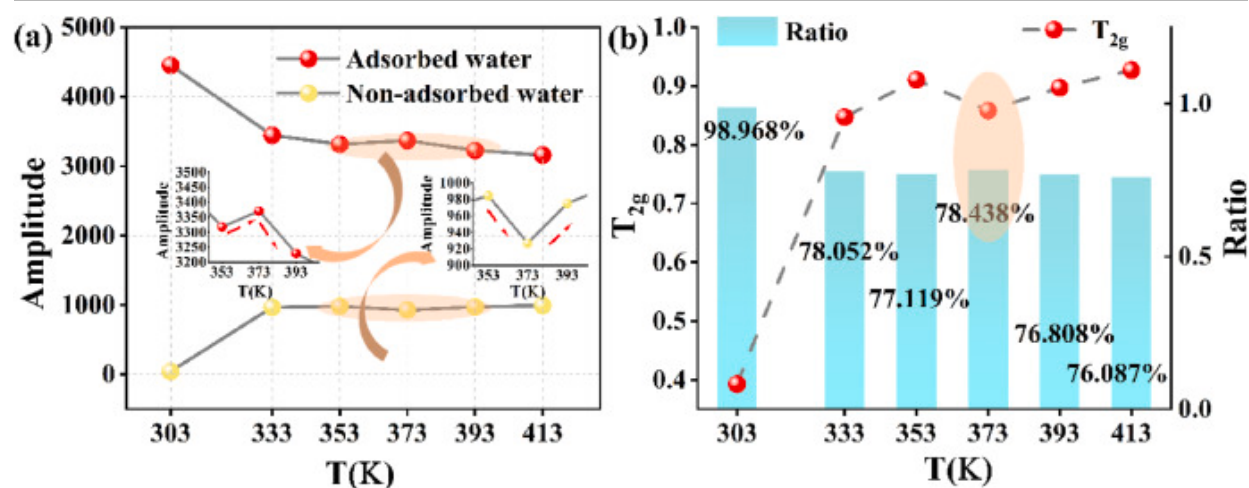


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Fig. 4. Variations in T₂ spectra at different CO₂ injection temperatures.

To better quantify the changes in the water wettability of coal after CO₂ injection at different temperatures, the total amplitude of adsorbed water, the total amplitude of non-adsorbed water, and T_{2g} after CO₂ injection were calculated. It is noteworthy that the T_{2g} values after CO₂ injection at different temperatures were obtained based on Eq. (3). As illustrated in Fig. 5(a), the trends in the total amplitudes of adsorbed water and non-adsorbed water correspond to each other, both exhibiting anomalies at 373.00K. To be specific, when the temperature rises from 353.00K to 373.00K, the total amplitude of adsorbed water grows from 3318.01 to 3371.39, while that of non-adsorbed water declines from 984.44 to 929.76. As depicted in Fig. 5(b), the T_{2g} value and the proportion of the two types of water also present different trends at 373.00K compared to other temperature rise stages. When the temperature rises from 353.00K to 373.00K, the proportion of adsorbed water in the coal no longer decreases. Instead, it increases from 77.11% to 78.43%. Meanwhile, T_{2g} no longer increases, but decreases from 0.91% to 0.85%. These results suggest that water in the coal migrates to smaller pores and are indicative of more remarkable water wettability.



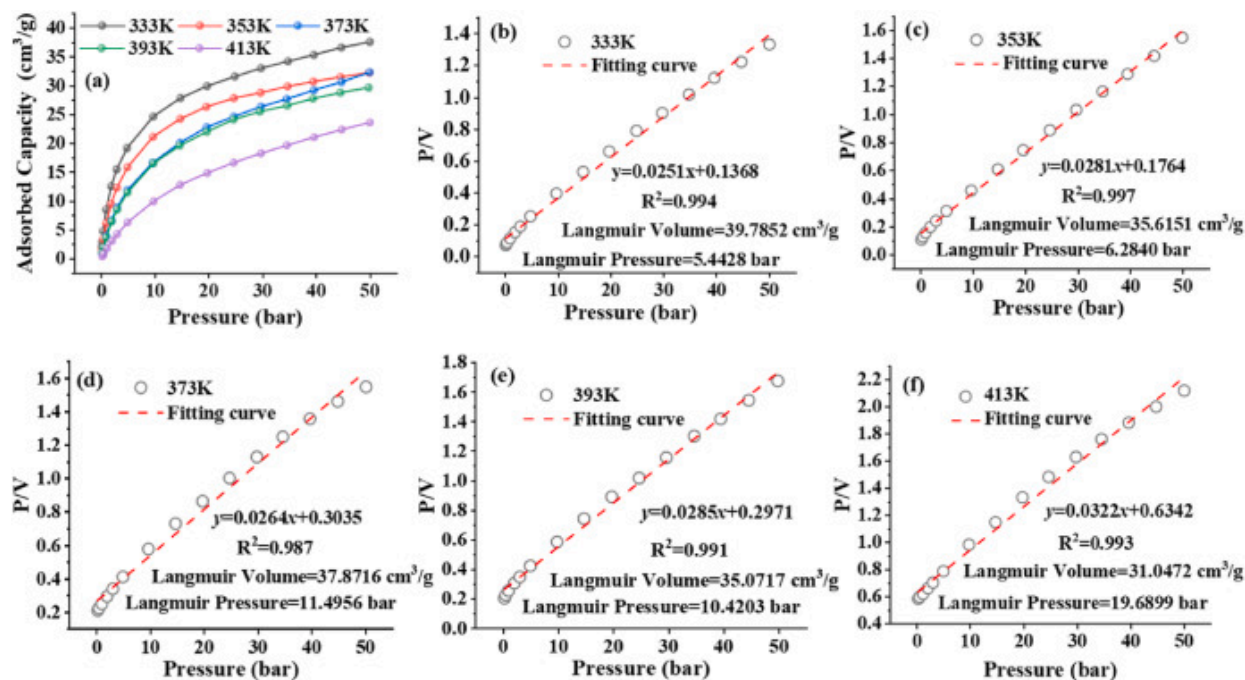
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Fig. 5. Content of adsorbed water and non-adsorbed water in pores at different CO₂ injection temperatures.

3.1.2. High temperature CO₂ adsorption on coal surface

In Fig. 6(a), the adsorption isotherms exhibit similar trends at different CO₂ injection temperatures, resembling Type I adsorption isotherms. In this study, we focused on investigating the changes in CO₂ adsorption characteristics in coal with the rise of CO₂ injection temperature at 4.00MPa. Overall, the sensitivity of CO₂ adsorption capacity to temperature at 4.00MPa changes in three stages (relatively high, low, and high). Specifically, when the injection temperature rises from 333.00K to 353.00K, the CO₂ adsorption capacity declines from 35.31 cm³/g to 30.69 cm³/g, by 4.62 cm³/g. Within the temperature range of 353.00–393.00K, the CO₂ adsorption capacity becomes less sensitive to temperature. Even at 393.00K, the CO₂ adsorption capacity remains as high as 27.76 cm³/g, that is, the reduction is merely 2.93 cm³/g over a temperature rise of 40.00K. Such a result unveils that CO₂ still possesses strong adsorption capacity in coal. It should be noted that the improvement in water wettability observed in the LF-NMR experiments with the rise of CO₂ injection temperature also occurs within this temperature range. As the CO₂ injection temperature continues to rise, the adsorption capacity becomes extremely sensitive to temperature, dropping to 21.07 cm³/g at 413.00K, with a reduction of 6.69 cm³/g.



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Fig. 6. Adsorption of CO₂ in coal at different CO₂ injection temperatures.

To better evaluate the changes in CO₂ adsorption characteristics in coal at different temperatures, the adsorption data were fitted according to the Langmuir adsorption curve, and the Langmuir volume and pressure were obtained from the slope and the intercept on the Y-axis:

$$\frac{p}{V} = \frac{p}{V_L} + \frac{p_L}{V_L} \quad (7)$$

where p is the gas pressure, bar; V is the adsorption capacity at pressure p , cm³/g; V_L is the Langmuir volume, cm³/g; and p_L is the Langmuir pressure, bar. As depicted in Fig. 6(b)–(f), the V_L value is 39.78 cm³/g, 35.61 cm³/g, 37.87 cm³/g, 35.07 cm³/g, and 31.04 cm³/g at 333.00K, 353.00K, 373.00K, 393.00K, and 413.00K, respectively. The corresponding p_L

values are 5.44bar, 6.28bar, 11.49bar, 10.42bar, and 19.68bar. It is evident that, except for the case at 373.00K, the V_L value continuously decreases with the rise of temperature, indicating a reduction in the ultimate adsorption capacity of coal for CO₂. Meanwhile, p_L noticeably increases at injection temperatures of 373.00K and 413.00K, with increases of 5.21bar and 9.26bar compared to the values at 353.00K and 393.00K, respectively. Thus, at 373.00K, although CO₂ possesses a stronger adsorption capacity in coal than at 353.00K, it requires much higher adsorption driving force (even 1.07bar higher than the value at 353.00K). At 413.00K, p_L reaches the maximum value in this study, and intense molecular thermal motion makes it difficult for CO₂ molecules to adsorb onto coal. Considering both V_L and p_L results, the ultimate adsorption capacity of coal for CO₂ barely changes within the low-sensitivity range (353.00–393.00K), whereas the adsorption capacity of CO₂ is dramatically enhanced at relatively lower temperatures and low pressures. By comparing the results with those from the LF-NMR experiments, it is found that under the conditions of this study (a CO₂ injection pressure of 4.00MPa), the adsorption capacity of CO₂ in coal does not decrease at the injection temperature where the water wettability exhibits anomalies (373.00K). In fact, it even surpasses the ultimate adsorption capacity at the lower temperature of 353.00K. Furthermore, the injection pressure of 4.00MPa is far higher than p_L at 373.00K, which provides sufficient adsorption driving force for CO₂ in coal.

3.1.3. Analysis of three-phase contact angle

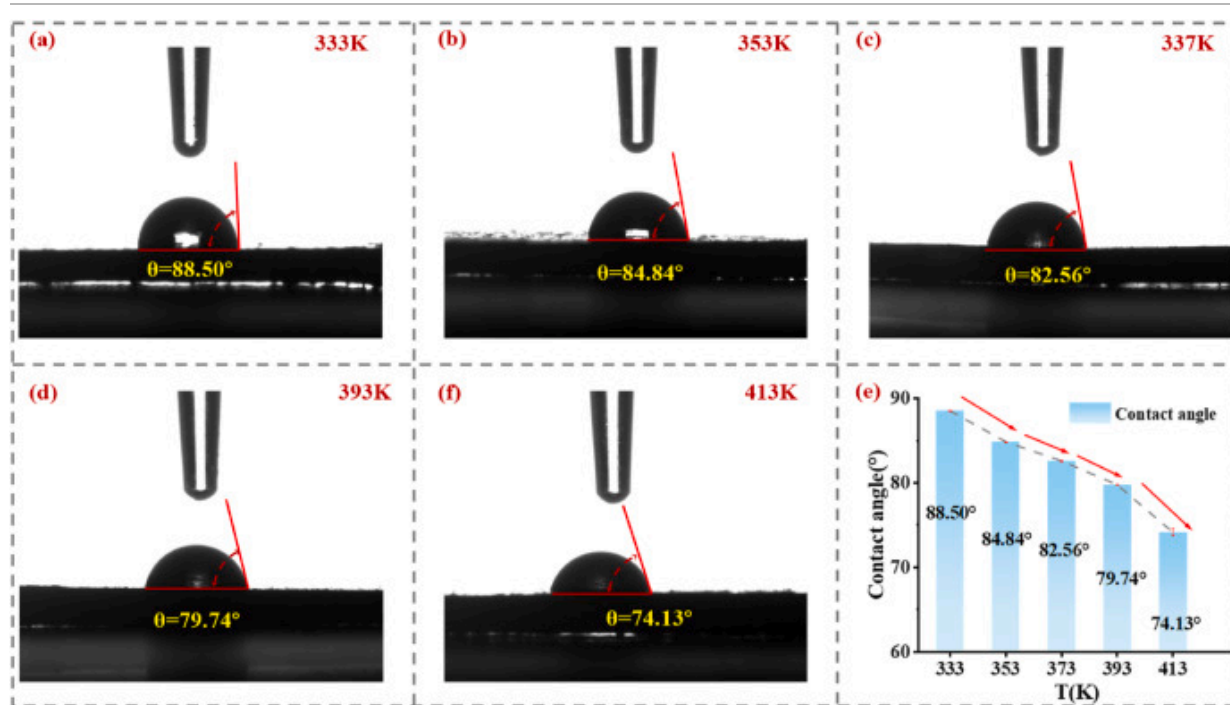
The contact angle measurement results at different temperatures are shown in [Table 2](#).

Table 2. Contact angle data.

T(K)	Contact angle(°)				
	Test No.1	Test No.2	Test No.3	Average	Standard deviation
333K	88.4	88.6	88.5	88.5	0.1
353K	84.92	84.71	84.89	84.84	0.11358

T(K)	Contact angle(°)				
	Test No.1	Test No.2	Test No.3	Average	Standard deviation
373K	82.52	82.63	82.53	82.56	0.06083
393K	79.84	79.89	79.49	79.74	0.21794
413k	74.17	73.94	74.28	74.13	0.17349

As shown in Fig. 7(a)–(f), as the CO₂ injection temperature rises from 333.00K to 413.00K, the three-phase contact angle (θ) value continuously shrinks, being 88.50°, 84.84°, 82.56°, 79.74°, and 74.13°, respectively. The decrement in θ shows a trend of first increasing, then decreasing, and finally increasing again. When the CO₂ injection temperature rises from 333.00K to 353.00K, the decrement is 3.66°. As the temperature further rises to the CO₂ low-sensitivity zone identified in the above analysis (353.00–393.00K), a 40K increase in CO₂ injection temperature only corresponds to a 5.1° decrement in θ . However, when the CO₂ injection temperature further rises to 413.00K, a 20K increase in CO₂ injection temperature corresponds to a 5.61° decrease in θ . It is evident that the variation trend of θ with the rise of CO₂ injection temperature is closely related to the adsorption capacity of CO₂ in coal.

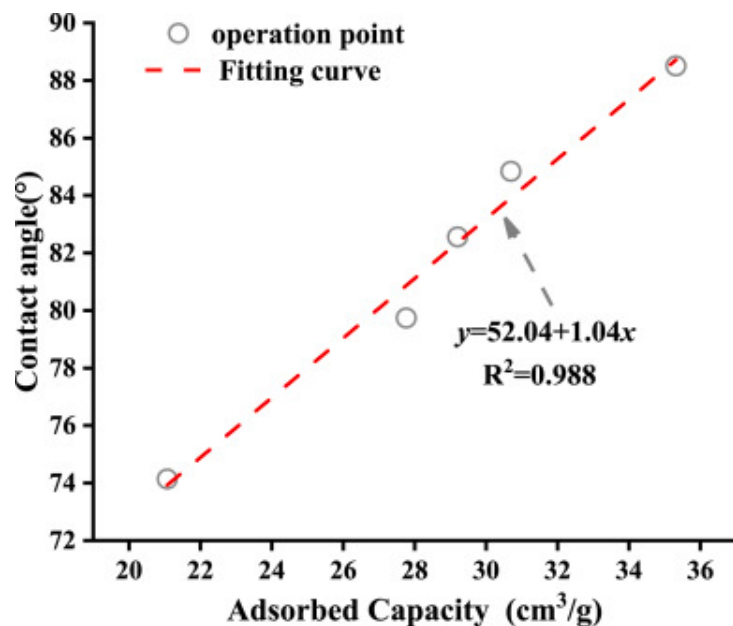


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Fig. 7. Changes in θ at different CO₂ injection temperatures.

To better investigate the relationship between θ and CO₂ adsorption capacity, a scatter plot was created by taking the adsorption capacities at different injection temperatures as the X-axis and the θ values as the Y-axis, and curve fitting was conducted. As shown in Fig. 8, the adsorption capacity of CO₂ is linearly correlated with θ , that is, θ continuously grows as the adsorption capacity rises.



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Fig. 8. Relationship between CO₂ adsorption capacity and θ .

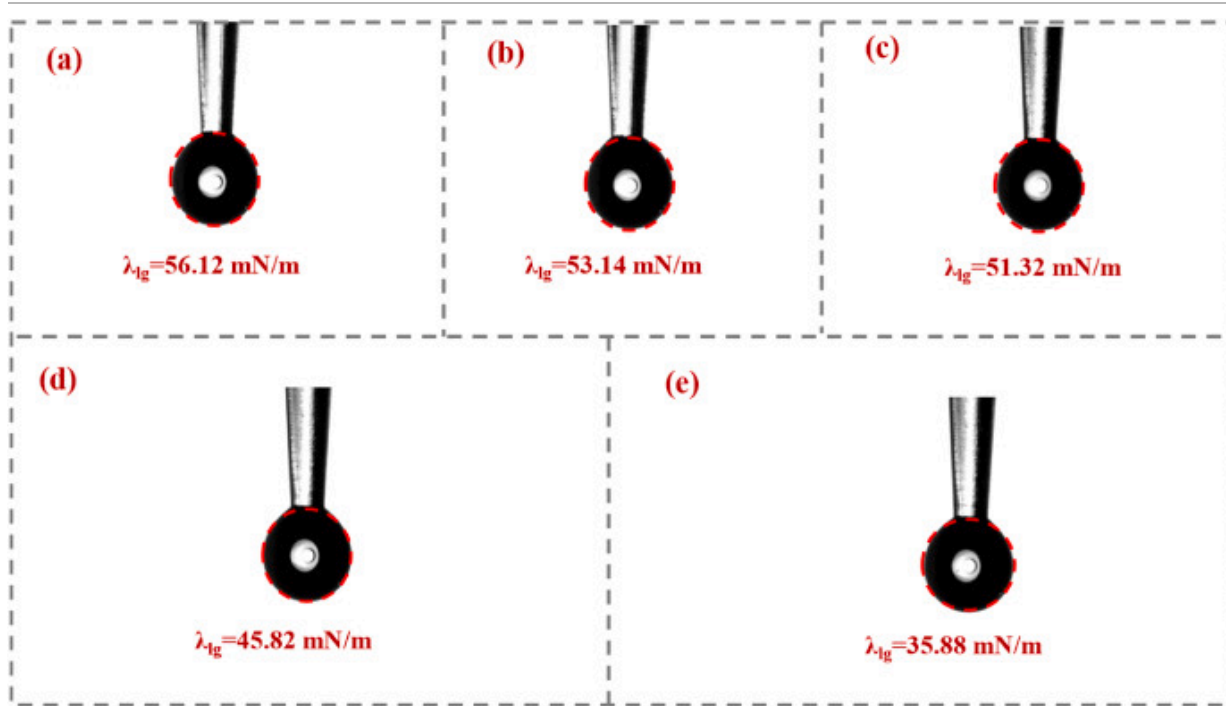
3.1.4. Changes in key parameters of wetting characteristics at the interface

To comprehensively evaluate the influence of high-temperature CO₂ injection on the water wettability of coal, the surface tension of droplets was tested in a CO₂ atmosphere at 4.00MPa and different CO₂ injection temperatures. The surface tension of the gas-water interface at different temperatures is shown in [Table 3](#).

Table 3. Surface tension data.

T(K)	Surface tension(mN/m)				
	Test No.1	Test No.2	Test No.3	Average	Standard deviation
333K	56.12	56.14	56.1	56.12	0.02
353K	53.21	53.09	53.12	53.14	0.06245
373K	51.36	51.29	51.31	51.32	0.03606
393K	45.83	45.75	45.88	45.82	0.06557
413K	35.96	35.87	35.81	35.88	0.0755

All data in this study are the average values of multiple tests conducted at the same temperature, as shown in Fig. 9. As the CO₂ injection temperature rises, the thermal motion of CO₂ molecules outside the droplets intensifies, and the attraction between gas molecules and liquid molecules is enhanced. This reduces the uneven force distribution on water molecules at the gas-liquid interface, leading to a continuous decrease in gas-liquid interfacial tension from 56.12mN/m at 333.00K to 35.88mN/m at 413.00K. In the meantime, the inward clustering of water molecules on the droplet surface slows down. However, a decrease in the gas-liquid interfacial tension does not necessarily marks an enhancement in the wettability of droplets on the coal surface. In a high-temperature and high-pressure CO₂ atmosphere, the change in the wettability of droplets is also related to the coal-water interfacial energy λ_{sl} (coal-water interfacial tension), the surface free energy of coal λ_{sg} (coal-CO₂ surface tension), and the adhesion work W_a (see Fig. 9).



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Fig. 9. Gas-liquid interfacial tension at different CO₂ injection temperatures.

The coal-water interfacial energy λ_{sl} , surface free energy of coal λ_{sg} , and adhesion work W_a were obtained using Eqs. (8), (9), (10), respectively:

$$\lambda_{sl} = \frac{\lambda_{lg}}{2} \left(\sqrt{1 + \sin^2 \theta} - \cos \theta \right) \quad (8)$$

$$\lambda_{sg} = \frac{\lambda_{lg}}{2} \left(\sqrt{1 + \sin^2 \theta} + \cos \theta \right) \quad (9)$$

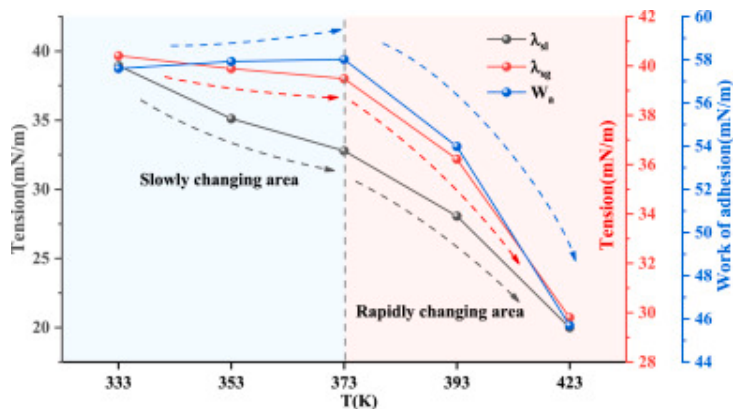
$$W = \lambda_{lg} (1 + \cos \theta) \quad (10)$$

where γ_{lg} is the surface tension of water in a CO₂ atmosphere, mN/m; θ is the contact

angle, °; γ_{sl} is the coal-water interfacial energy, mN/m; γ_{sg} is the surface energy of coal, mN/m; and W_a is the adhesion work, mN/m.

As displayed in Fig. 10, in a CO₂ atmosphere, the changes in the coal-water interfacial characteristic parameters are divided at 373.00K. When the CO₂ injection temperature is below 373.00K, the coal-water interfacial characteristic parameters vary slowly with temperature. In contrast, when the temperature exceeds 373.00K, these parameters change more violently with temperature. γ_{sg} is related to the potential for surface water molecules to wet the coal surface. Coal with a higher γ_{sg} tends to reduce its surface energy and achieves a stable configuration by allowing water molecules to wet its surface. Therefore, a higher γ_{sg} means that the coal can be wetted by water more easily. Compared to the cases at 333.00K and 353.00K, when the CO₂ atmosphere temperature rises to 373.00K, γ_{sg} declines slightly from 40.41 mN/m to 39.48 mN/m, by less than 1 mN/m. The coal surface still has abundant energy and a large potential to be wetted by water. However, once the CO₂ atmosphere temperature exceeds 373.00K, γ_{sg} drops markedly with the rise of temperature, with surface energy being only 36.22 mN/m and 29.79 mN/m at 393.00K and 413.00K, respectively, marking weakened water wettability. W_a , which is defined as the work required to separate two phases, is an important indicator for measuring wettability. The greater the γ_{sg} , the better the water wettability of coal. Notably, before the critical temperature of 373.00K, as the CO₂ atmosphere temperature increases, the γ_{sg} goes upward slightly, from 57.58 mN/m at 333.00K to 58.01 mN/m at 373.00K, demonstrating enhanced water wettability. Similar to γ_{sg} , when the CO₂ atmosphere temperature continues to rise to temperatures above 373.00K, W_a plunges, which implies that the solid and liquid phases separate more easily and the water wettability worsens significantly. A smaller γ_{sl} means that droplets spread on the coal surface more easily, corresponding to stronger water wettability. γ_{sl} presents a downward trend with the rise of temperature at all temperatures involved in this study. From the perspective of γ_{sl} alone, the coal surface is considered to achieve the best wettability when the CO₂ atmosphere temperature reaches the highest value. Nevertheless, under the comprehensive consideration of γ_{sg} , W_a , and γ_{sl} , it can be concluded that when the

injection temperature is 373.00K, the combination of the highest W_a , a γ_{sg} nearly the same as the value at 353.00K, and a relatively small γ_{sl} results in the best water wettability. Once the temperature exceeds 373.00K, the sharp decreases in W_a and γ_{sg} considerably worsen the water wettability. When the temperature is below 373.00K, the relatively large γ_{sl} limits the spreading of droplets on the coal surface, leading to poor water wettability. These observations well explain the occurrence of the abnormal wettability phenomenon at 373.00K in LF-NMR experiments.



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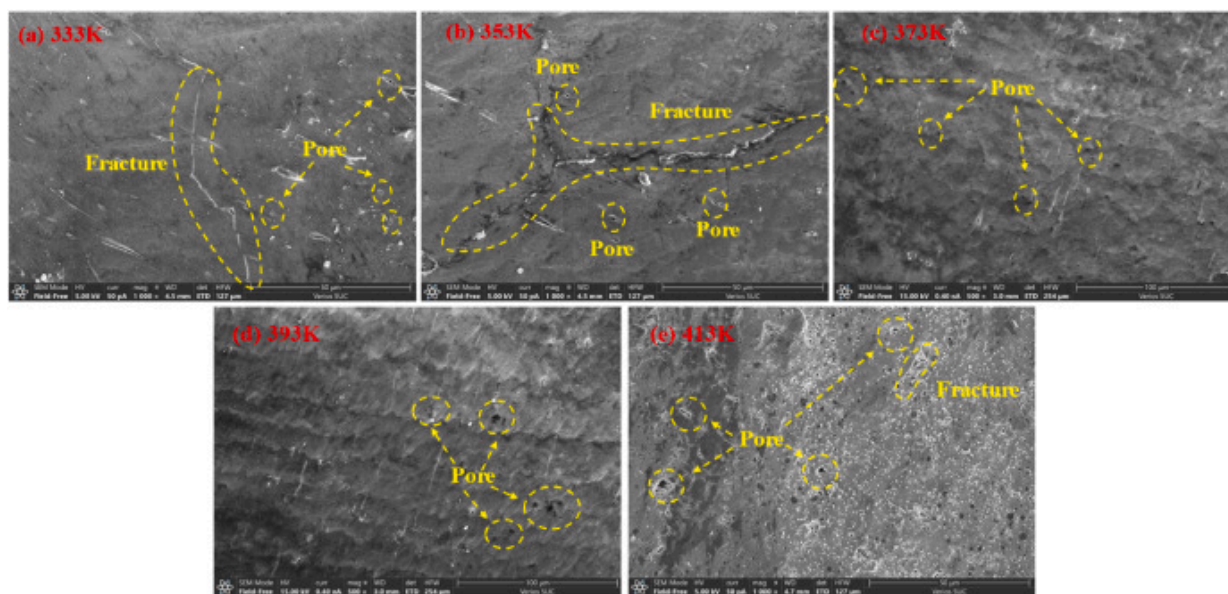
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Fig. 10. Coal-water interfacial wettability characteristic parameters at different CO₂ atmosphere temperatures.

3.1.5. Surface characteristics of coal after CO₂ injection at different temperatures

The SEM images of the surface of water containing coal after CO₂ treatment at different temperatures are shown in [Fig. 11](#). The anthracite utilized in this study exhibits a dense texture. After CO₂ treatment at varying temperatures, only a limited number of pores and fractures are observed on the surface, with their edges remaining relatively smooth. No significant pore expansion, fracture propagation, or noticeable generation of new pores and fractures is detected. Consequently, the influence of CO₂ on the surface characteristics

of the coal is found to be negligible in this study. This is attributed to the fact that in this study, most moisture in the coal matrix exists as adsorbed water within pores and fractures, exhibiting relatively low water content. At elevated temperatures, the reduced solubility of CO₂ in water leads to only trace formation of carbonic acid, which is inadequate to effectively corrode surface minerals and modify the pore-fracture structure. Moreover, since CO₂ remains in the gaseous phase throughout this study, its influence on coal surface characteristics is considerably weaker compared to that in the supercritical state.



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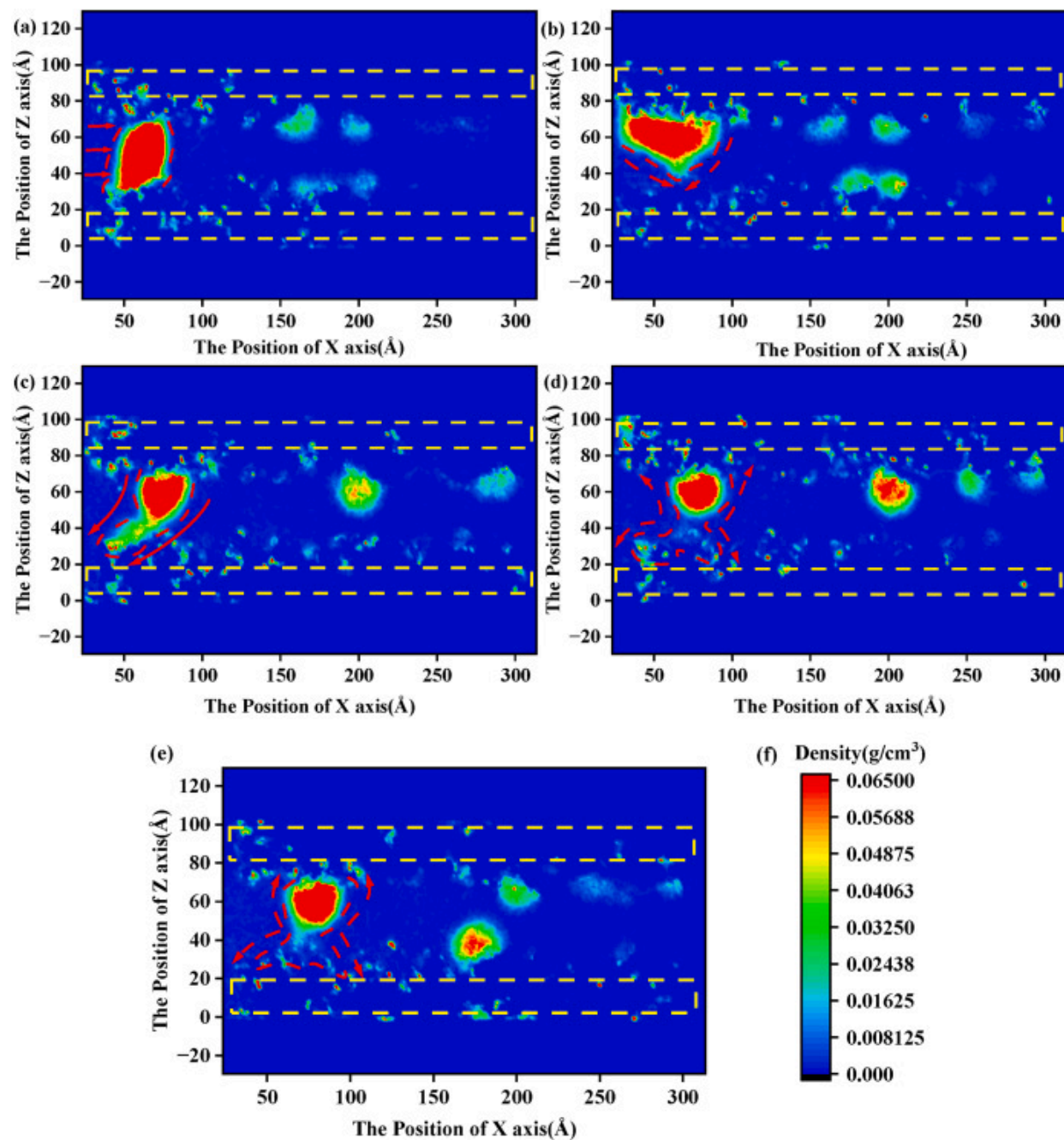
Fig. 11. SEM micrographs of coal surfaces treated with CO₂ at different temperatures (a) 333.00K (b) 353.00K (c) 373.00K (d) 393.00K (e) 413.00K.

3.2. Analysis on molecular dynamics simulation results

3.2.1. Density contour plots of water distribution in nano-slits after injection

Fig. 12(a)–(e) present the contour plots of H₂O density at CO₂ injection temperatures of 333.00K, 353.00K, 373.00K, 393.00K, and 413.00K, respectively. Clearly, the injection of high-temperature CO₂ has different effects on water clusters adsorbed in large volumes on the upper and lower walls of micro nano-slits within the coal and water molecules occurring in small volumes within the deep coal matrix. For the former, the activation of water molecules caused by the temperature field arising from the injection of high-temperature CO₂ has a stronger impact on wettability than the competitive wetting of the two types of molecules within the coal. This activation effect strengthens with the rise of CO₂ injection temperature. When the CO₂ injection temperature is 333.00K, the water clusters at the left entrance of the slit change from a hemispherical to an ellipsoidal-like shape under the disturbance of CO₂ molecules, indicating poorer water wettability. However, the low-density shell formed by the increased internal energy of water molecules is relatively thin, so the change in wettability of water clusters is mainly induced by the disturbance of the high-pressure gas. When the CO₂ injection temperature rises to 353.00K, more outer water molecules are activated by heat, and their thermal motion becomes more intense. Meanwhile, the low-density shell becomes much thicker. An unstable “peak” configuration appears at the top of the nano-water clusters, and the outer water molecules tend to detach. When the temperature further rises to 373.00K, the intermolecular hydrogen bonds can no longer restrict the movement of water molecules in the low-density shell. Consequently, some outer water molecules detach from the nano-water clusters and move within a larger pore space. This promotes the opportunity for water molecules to contact the coal surface and makes water molecules more likely to be captured by the slit walls. As a result, water molecules detaching from the nano-water clusters in Fig. 12(c) are “re-absorbed” onto the lower wall of the slit. The continuous detachment of outer water molecules reduces the volume of the nano-water clusters, causing them to move inward within the slit under the disturbance of the high-pressure gas. When the temperature reaches 393.00K and 413.00K, the activated water molecules

have excessively high temperature and internal energy, and their intense thermal motion prevents them from being captured by adsorption sites on the coal surface. Carried by the high-pressure gas, they quickly separate from the nano-water clusters and migrate out of the slit space.



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Fig. 12. Contour plots of H₂O density (a) 333.00K (b) 353.00K (c) 373.00K (d) 393.00K (e) 413.00K (f) Legend.

The density of water molecules stored in the deep coal matrix exhibits a three-stage characteristic of first dropping, then increasing, and finally gradually declining as the CO₂ injection temperature rises. Besides, areas with high CO₂ adsorption densities appear in the slit entrance region where the value on the X-axis is smaller than 100.00Å. Since gases have poor thermal conductivity, the thermal effect brought about by the injection of high-temperature CO₂ plunges along the Y-axis as it penetrates deeper into the coal matrix, failing to exerting a strong overall heating effect on the coal matrix. The movement of CO₂ molecules that are captured by the slit walls and enter the coal matrix is highly restricted, followed by a drop of their temperature and an enhancement of their adsorption capacity within the complex pore network of the anthracite matrix. Water molecules stored in the deep coal matrix are characterized by their small volume and few cluster molecules. These characteristics are conducive to competitive wetting by CO₂ molecules. A comparison between Fig. 12, Fig. 17 reveals that areas with high CO₂ adsorption densities in the deep coal matrix witness sharp decreases in the adsorption density of water molecules, particularly in the latter half of the slit where the value on the X-axis is greater than 200.00Å.

3.2.2. RDF curves

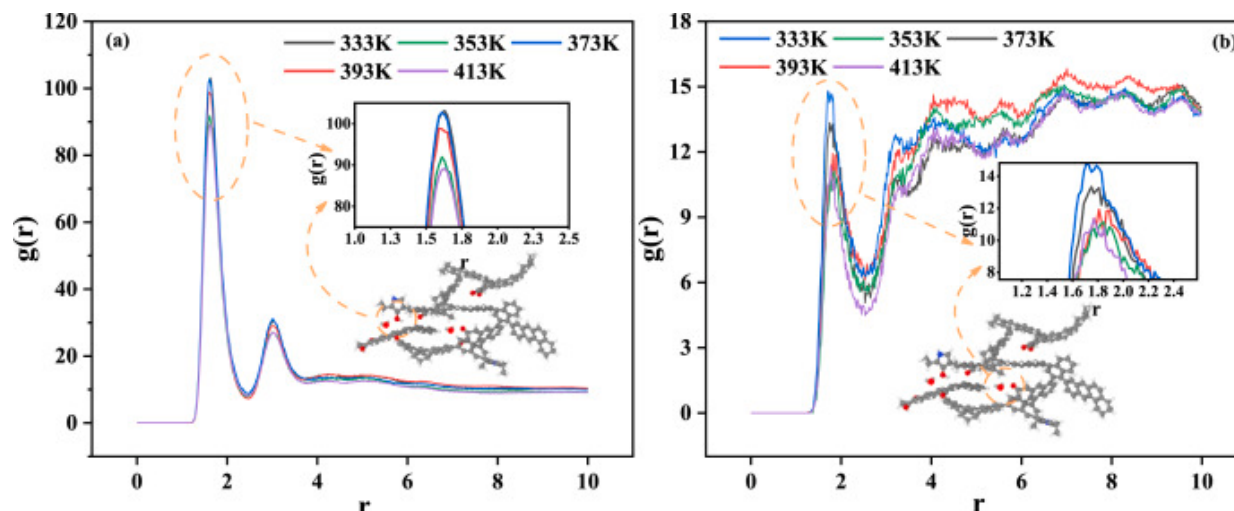
Oxygen-containing functional groups serve as crucial sites for water molecule adsorption onto coal, and the distribution density of water molecules distributed around these groups greatly influences the changes in water wettability of coal during CO₂ injection at different temperatures. Based on the constructed macromolecular structural characteristics of anthracite, this study investigates the variation in the radial distribution function (RDF) between oxygen atoms in two typical oxygen-containing functional groups (carbonyl and hydroxyl) and hydrogen atoms in water molecules with the rise of CO₂ injection temperature (Niu et al., 2025). In the rough slit model constructed in this study, hydroxyl groups are located closer to the slit surface area compared to carbonyl groups, so

these two groups can represent the distribution of water molecules on the slit wall and within the matrix, respectively. The RDF is obtained by Eq. (11):

$$g(r) = \frac{dN}{\rho 4\pi r^2 dr} \quad (11)$$

where r is the distance from the central atom; ρ is the density of the system; and dN is the number of atoms within the range that is from r to $r+dr$ away from the central atom.

As shown in Fig. 13(a), distinct peaks appear at 1.50Å and 3.00Å on the RDF curves, indicating the presence of two high-density adsorption regions for water molecules on the slit walls. This finding aligns well with the double-layer adsorption pattern discovered by scholars. The primary peak which is formed by inter-molecular hydrogen bonds between oxygen atoms in hydroxyl groups and hydrogen atoms in water molecules holds a much higher intensity than the secondary peak which results from hydrogen bonds between water molecules (Dannenberg, 1998). As the CO₂ injection temperature rises from 333.00K to 413.00K, the intensity of the primary peak drops initially, goes up subsequently, and declines again finally. Such a three-stage variation pattern is consistent with the variation in adsorbed water observed in LF-NMR experiments. From the perspective of inter-molecular interactions, the above analysis verifies the detachment and “re-adsorption” of water molecules in adsorption clusters on the slit walls, as well as their migration from larger-scale flow paths (fractures) to smaller-scale storage spaces (nanopores).



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Fig. 13. RDF curves of different O atoms in coal and H atoms in H₂O.

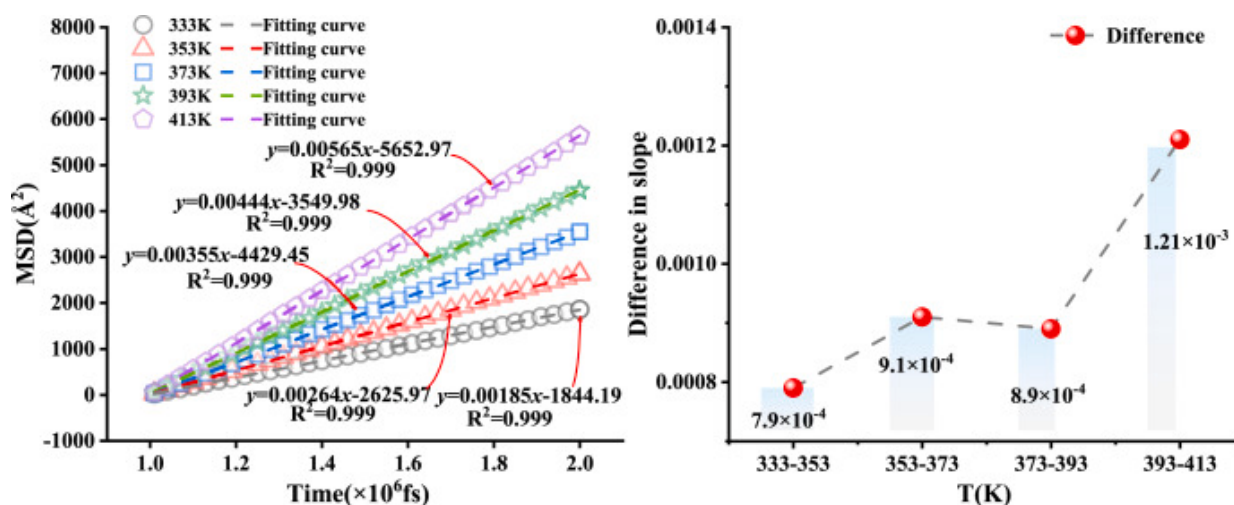
As depicted in Fig. 13(b), the RDF curves of O atoms in carbonyl groups and H atoms in water presents only one prominent peak, but the $g(r)$ value remains high after 5 Å. This is related to the spatial position of carbonyl groups within the coal matrix. Specifically, multiple adsorption regions are in close proximity within the coal matrix, thereby inducing a certain degree of overlap. It is noteworthy that, although the adsorption density of water molecules near carbonyl groups at 373.00K is far higher than those at 353.00K, 393.00K, and 413.00K, it is still lower than that at 333.00K. The result implies that the migration of water molecules from the slit walls into the matrix interior is more challenging compared to their detachment from clusters and adsorption onto the slit walls.

3.2.3. Mean squared displacement and diffusion coefficient of water molecules

The mean squared displacement (MSD) of water molecules at different CO₂ injection temperatures was calculated in the hope of exploring the activity level of pre-existing water within the nano-slits and the coal matrix in a high-temperature and high-pressure CO₂ environment (Fig. 14) (Gan et al., 2023). At different CO₂ injection temperatures, the MSD is linearly correlated with the simulation time, and one-sixth of the slope represents the diffusion coefficient of water molecules. The MSD and diffusion coefficient are obtained through Eq. (12) (Hehe et al., 2022; Jin et al., 2022; Li et al., 2022):

$$D = \frac{1}{6N} \lim_{t \rightarrow \infty} \frac{d}{dt} \sum_{i=1} \langle [r_i(t) - r_i(0)]^2 \rangle \quad (12)$$

where N is the number of water molecules; $r_i(t)$ and $r_i(0)$ are the position vectors of atoms at time t and the initial moment, respectively.



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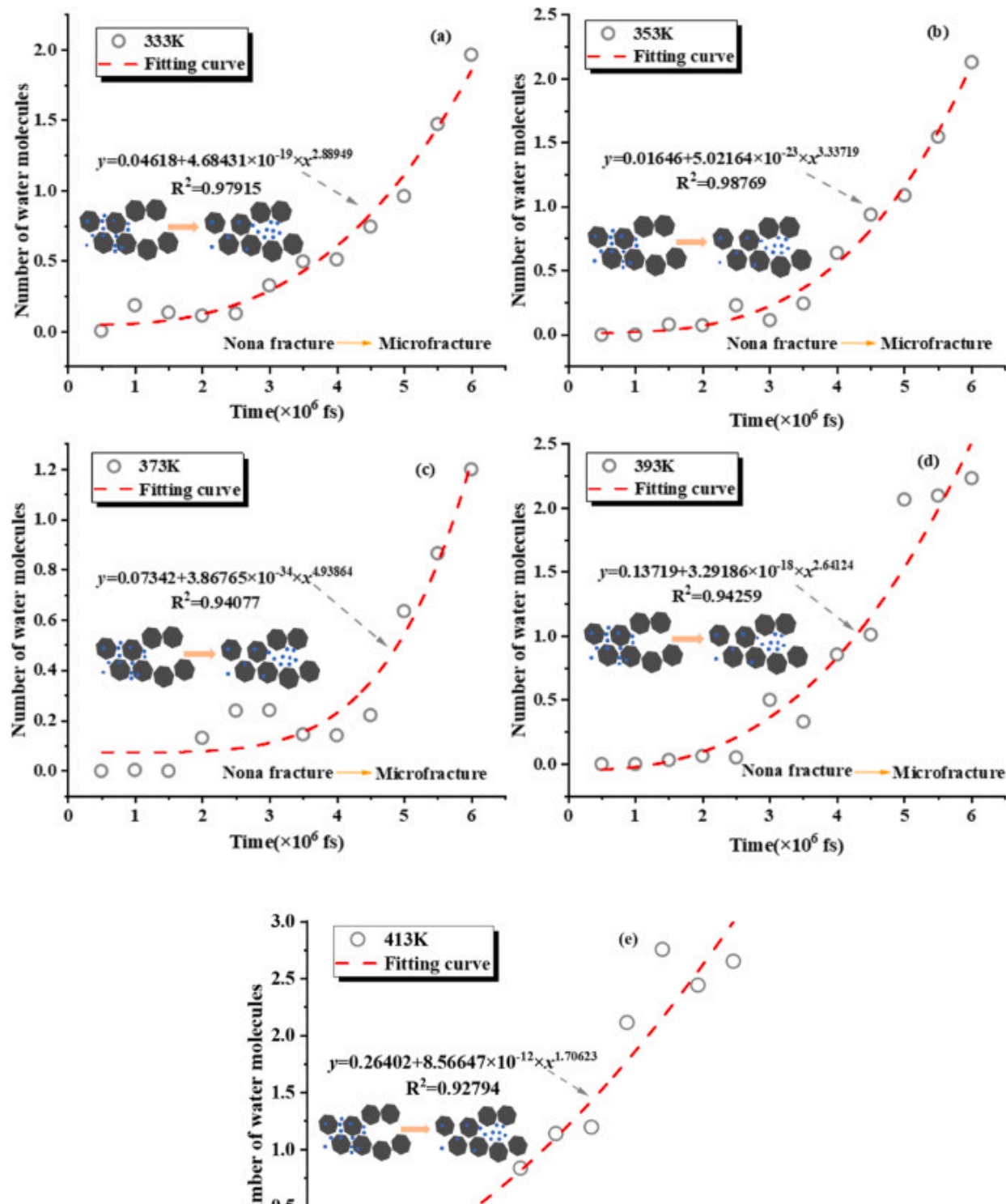
Fig. 14. Activity of H₂O molecules in nanopores of coal.

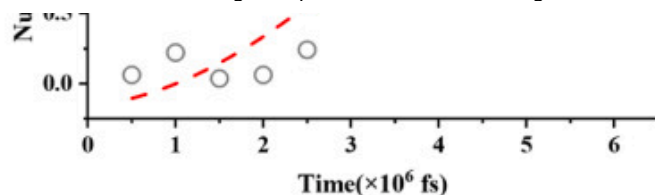
When the CO₂ injection temperature is only 333.00K, the diffusion coefficient of pre-existing water within both the nano-slits and the coal matrix is relatively small, just 0.0003 Å²/fs, demonstrating a limited range and intensity of thermal motion for water

molecules. For water molecule clusters adsorbed on the nano-slit surfaces, due to strong intermolecular hydrogen bonding, water molecules prefer multilayer adsorption rather than spreading over a larger area on the slit surfaces, which corresponds to the larger three-phase contact angle and gas-liquid surface tension at the macroscopic level. For confined water molecules within the coal matrix, the lower diffusion coefficient favors the stability of their adsorption configuration, which is consistent with the high density of water molecules in the coal matrix observed in Fig. 12(a). As the CO₂ injection temperature rises to 353.00K, the diffusion coefficient of water molecules increases to 0.00044Å²/fs, by only 0.00014Å²/fs. The small increment only notably affects the low-density shell outside large clusters adsorbed on the slit surfaces. The decrease in the density of adsorbed water within the coal matrix observed in Fig. 12(b) is closely related to the strong adsorption capacity of CO₂ at this temperature. As the CO₂ injection temperature further rises to 373.00K, the diffusion coefficient, 0.00074Å²/fs, becomes sufficient to detach some water molecules from the low-density shell outside the clusters without imparting excessive energy that would prevent their re-adsorption onto the slit surfaces and subsequent entry into the coal matrix. It is worth noting the increase in the diffusion coefficient of water molecules when the CO₂ injection temperature rises from 393.00K to 413.00K spans an order of magnitude, compared to the case when the CO₂ injection temperature rises from 373.00K to 393.00K. These results suggest that when the CO₂ injection temperature reaches 373.00K, a further slight increase in temperature can hardly significantly affect the diffusion behavior of water molecules and fail to provide sufficient potential for the continued intensification of molecular thermal motion. As a consequence, a threshold similar to an “energy barrier” emerges near this temperature. A substantial increase in temperature is required to effectively overcome this “energy barrier”. From the perspective of water molecules themselves, this “energy barrier” phenomenon explains why the density of water molecules on the slit surfaces and within the coal matrix grows at 373.00K in Fig. 12(c).

3.2.4. Migration characteristics of water molecules in multi-scale nanopores

In order to more intuitively demonstrate the migration of water molecules in nanopores, fractures, and micro-fractures in LF-NMR experiments at the molecular level and probe into changes in water wettability of coal under high-temperature CO₂ injection from a microscopic perspective, the number of water molecules at the right-end outlet of the slit and within the coal matrix was monitored in real time during the injection process (Fig. 15). The number of water molecules in the coal matrix can effectively represent the adsorbed water molecules within nanopores, while the change in the number of water molecules at the slit outlet is closely related to the migration of water molecules from nano-fractures to relatively larger micro-fractures.





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Fig. 15. Migration of H₂O molecules in different-scale nanopores of coal.

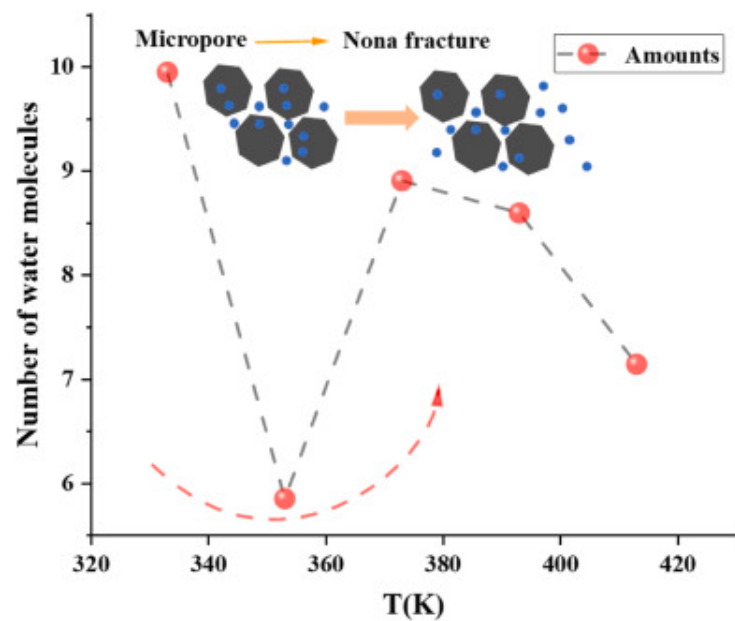
As can be found from Fig. 15(a)–(e), the number of water molecules migrating from nano-fractures to relatively larger micro-fractures during CO₂ injection at different temperatures exhibits a power function relationship (in which an offset constant c is incorporated) with the simulation time. The expression is as follows:

$$y = a \times x^b + c \quad (13)$$

where c is the offset constant; a is the amplitude factor; b is the power exponent. In this study, the offset constant c allows the data to maintain high fitting accuracy in the presence of baseline drift or offset, although its confidence interval is not significant. When analyzing the variation trend of the function, emphasis should be placed on the changes in parameters a and b at different temperatures. Before the transition temperature, 373.00K, as the CO₂ injection temperature rises, the number of water molecules migrating from nano-fractures to relatively large micro-fractures grows slowly in the initial stage and soars in the later stage ($t > 3$ ns). During this heating process (333.00–373.00K), a continuously drops from 4.68×10^{-19} at 333.00K to 3.86×10^{-34} at 373.00K, while b constantly increases from 2.88 at 333.00K to 4.94 at 373.00K. A smaller a and a larger b result in an overall lower but faster rising curve, which is unfavorable for the transition from water-wet to gas-wet conditions. After the transition temperature, 373.00K, as the CO₂ injection temperature continues to rise, the number of water molecules grows more uniformly. During this heating process (373.00–413.00K), a keeps going up from 3.86×10^{-34} at 373.00K to 8.56×10^{-12} at 413.00K, while b keeps going down

from 4.94 at 373.00K to 1.70 at 413.00K. Moreover, as the CO₂ injection temperature rises, the inflection point at which the number of water molecules at the slit outlet soars occurs at an earlier stage of the simulation. It advances from 4.50 ns at 373.00K to 2.50 ns at 413.00K. This indicates that during the heating process after the transition temperature of 373.00K, a growing number of water molecules migrate constantly from nano-fractures to relatively large micro-fractures throughout the simulation, leading to a considerable deterioration in water wettability.

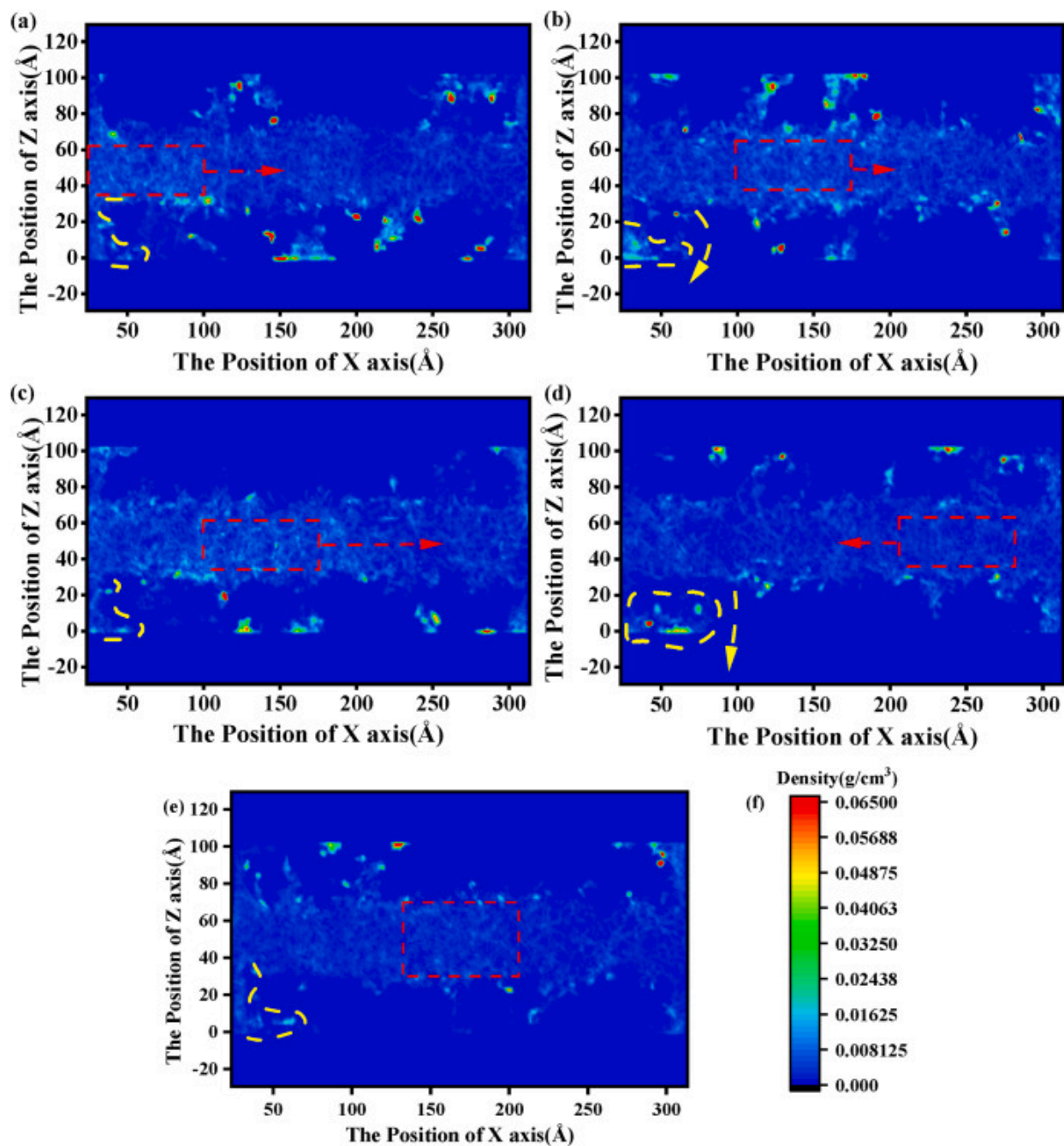
The variation of the time-averaged number of adsorbed water molecules in the nanopores of the coal matrix with the rise of CO₂ injection temperature is given in [Fig. 16](#). Clearly, it exhibits a three-stage pattern of an initial drop, a subsequent rise, and a final decrease, with 373.00K being the turning point. When the CO₂ injection temperature rises from 333.00K to 353.00K, the number of water molecules in the nanopores drops most dramatically, from 9.95 to 5.85. As the temperature continues to rise to 373.00K, influenced by factors such as changes in the energy of water molecules themselves, functional groups on the coal surface, and adsorption properties of CO₂, the number of water molecules in the nanopores returns to 8.90, indicating improved water wettability. When the temperature exceeds 373.00K, the number of water molecules in the nanopores decreases again, albeit at a slower rate. At 413.00K, a certain amount of moisture is still present in the micropores and nanopores. It can be seen that the migration pattern of water molecules in the multi-scale channels of coal obtained through molecular simulation is highly consistent with the LF-NMR experimental results.



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Fig. 16. Migration of H₂O molecules in multi-scale nanopores of coal.



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Fig. 17. Contour plot of CO₂ density.

3.2.5. Density distribution of CO₂ in nanopores after injection

CO₂ in the slit space exists mainly in two forms: 1) surface adsorption within the nano-slit and 2) confined adsorption in pores within the coal matrix. With respect to the former, unlike water molecules, CO₂ molecules do not show hydrogen bonding and are distributed in a planar manner within the pore channels. Besides, the density of CO₂ near the slit walls is higher than that in the free space within the slit. As shown in Fig. 17(a)–(b), when the CO₂ injection temperature rises from 333.00K to 353.00K, the dense CO₂ adsorption region within the slit (indicated by the red box) shifts approximately 70.00Å along the X-direction, and the difference in CO₂ density between the slit walls and the free space shrinks, which is attributed to the weakened adsorption capacity of CO₂ at a higher temperature. When the CO₂ injection temperature further rises to 373.00K (Fig. 17(c)), the position of the dense CO₂ adsorption region changes slightly, shifting by only about 20.00Å along the X-direction, and no remarkable difference in CO₂ density between the slit walls and the free space is observed. When the CO₂ injection temperature reaches 393.00K (Fig. 17(d)), the density of adsorbed CO₂ within the nano-slit surface plummets, and the dense CO₂ adsorption region disappears.

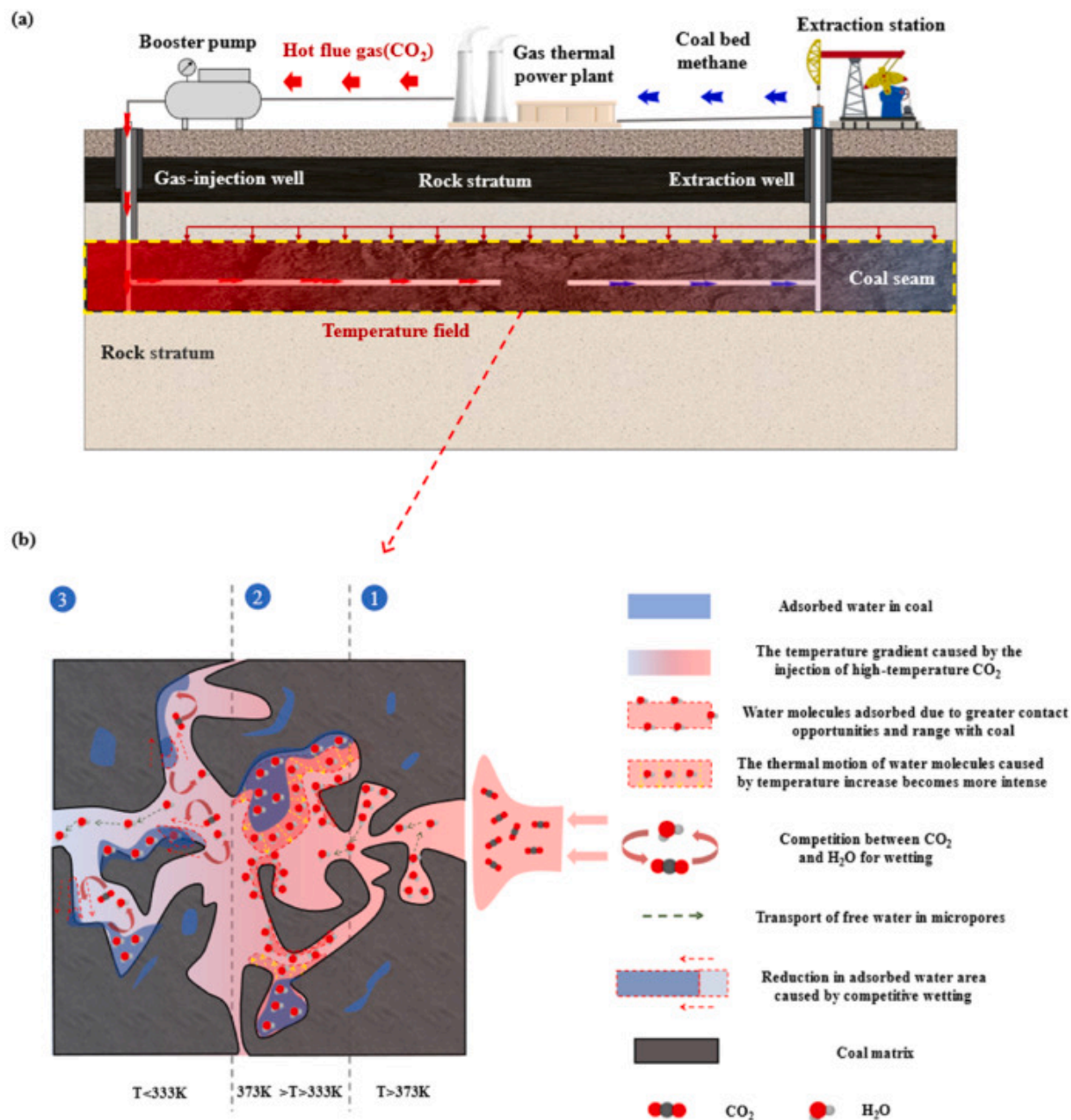
Some scholars have proposed that the presence of steady-state water films in porous materials conduces to CO₂ adsorption(Li et al., 2024). As can be observed from Fig. 9, apart from the large water clusters adsorbed on the upper and lower walls of the slit in a spherical-like manner, other water molecules are adsorbed on the upper and lower walls of the slit in the form of small aggregates, forming a structure resembling a discontinuous thin water film. The high-temperature CO₂ molecules dissolved in this film cool, accompanied by a reduction in the intensity of their thermal motion. Correspondingly, they are more likely to adsorb onto the slit walls or penetrate deeper into the coal matrix to become confined in an adsorbed state. Here is an analysis with a typical region at the entrance of the nano-slit as an example: When the CO₂ injection temperature is 333.00K, the weak molecular activation results in a relatively thin low-density shell on the outer

layer of the large water clusters, and the water molecule film on the lower wall of the slit is not integrated due to inter-molecular hydrogen bonding. The insufficient number of high-temperature CO₂ molecules dissolved in this film leads to a low density of confined CO₂ molecules adsorbed deep within the coal matrix. As the CO₂ injection temperature rises to 353.00K, the unstable configuration at the top of the nano-water clusters causes more water molecules to detach, forming a film with a more suitable thickness and a more complete structure on the lower wall. This film dissolves more CO₂ molecules, enabling them to contact the slit walls and move deeper into the coal matrix where they become confined in an adsorbed state. When CO₂ molecules with a relatively low initial temperature enter the coal matrix, their temperature further declines, which enhances their adsorption capability. They compete with water molecules to wet the coal surface, displacing water molecules from the nanopore space and reducing the density of water molecules in the coal matrix. This finding is in good agreement with the results presented in Fig. 9. As the injection temperature further increases to 373.00K (Fig. 17(c)), the “re-adsorption” phenomenon of water molecules further thickens the water film on the lower wall of the slit. The excessively high water content reduces the diffusion coefficient of CO₂ in the water film and hinders its contact with the slit walls. Resultantly, the density of adsorbed CO₂ in the coal matrix plummets. The increase in CO₂ adsorption density in the coal matrix at 393.00K occurs because the intense thermal motion of water molecules reduces the thickness of the water film, while the decrease at 413.00K arises from the fact that such an excessively high CO₂ injection temperature endows the matrix-entering molecules with high energy, which is unfavorable for adsorption.

4. Wettability variation pattern

Overall, the injection creates a temperature field (where the temperature decreases from the injection well toward the deeper parts of the coal seam) within the coal seam. When high-temperature CO₂ initially enters the coal seam, its high temperature creates a notable temperature difference with moisture in the coal, thus exerting a pronounced activation effect on water molecules adsorbed within the coal. As a result, the

temperature and internal energy of moisture increase, which leads to a rapid enhancement in its diffusion coefficient. Besides, the injected high-temperature CO₂ disrupts the initial adsorption configuration of moisture in the coal and the hydrogen bonds between water molecules. As a consequence, moisture migrates outward from the pore structure, with part of the adsorbed water transitioning into free water, which deteriorates the water wettability of the coal, as illustrated in Region ① in Fig. 18(b). It is noteworthy that the CO₂ adsorption capacity in the coal at high temperatures is found to be minimal in this study (at 413.00K, the CO₂ adsorption capacity is only 21.07cm³/g), which rules out the influence of CO₂-water competitive wetting on water wettability. This finding indicates that changes in wettability during this stage are mainly determined by the thermal field introduced by high-temperature CO₂.



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Fig. 18. Variation of water wettability in different areas of the coal body.

As CO₂ gradually migrates away from the injection well towards the interior of the coal, its temperature decreases. Correspondingly, the temperature difference with the moisture shrinks, and the activation effect on moisture in the coal weakens. Such a relatively weak activation effect would not cause moisture in the coal to immediately detach from its initial adsorption sites and migrate outside pores and fractures. Instead, it destabilizes the initial adsorption configuration of moisture in the coal and weakens the “binding” of hydrogen bonds between water molecules on the surface moisture. Resultantly, water becomes more mobile within pores and more likely to contact and adsorb onto other pore surfaces, as illustrated in Region ② in Fig. 18(b). In addition, as discussed in Section 3.1.2, when the CO₂ temperature lies within the range of 353.00–393.00K, the sensitivity of the CO₂ adsorption capacity to temperature plunges. With a temperature increase of 40.00K, the CO₂ adsorption capacity only declines by 2.93cm³/g, which implies that the improvement in water wettability is largely unrelated to the competitive wetting of CO₂. In conclusion, the moderate activation effect brought about by the smaller temperature difference between gas and water plays a dominant role in this stage, leading to enhanced water wettability of the coal.

As the CO₂ temperature drops further (333.00K), the activation effect on moisture arising from the temperature difference between gas and water becomes negligible. The slope of the mean square displacement of water molecules is merely 0.00185, indicative of low water mobility. On the other hand, the CO₂ adsorption capacity in coal surges in this stage, causing adsorbed water molecules to detach from adsorption sites and migrate outward under competitive wetting. Consequently, the water wettability weakens.

5. Conclusions

This study is aimed to uncover the patterns and mechanisms of water wettability changes in coal pores and fractures under high-temperature and high-pressure CO₂ injection (333.00–413.00K, 4.00MPa). To achieve this aim, both experiments and molecular

dynamics simulations were performed. The findings show three-stage water wettability changes with the increase of injection depth, characterized by a “decrease–increase–decrease” pattern.

- 1) When the CO₂ temperature remains high, its adsorption capacity in coal is minimal, 21.07 cm³/g. Meanwhile, the temperature field exerts a strong activation effect on water molecules adsorbed in the coal matrix, markedly promoting their internal energy and molecular mobility. At 413K, the slope of MSD of water molecules reaches 0.00565; the coal-water adhesion work and the surface energy of coal drop to 45.69 mN/m and 29.79 mN/m, respectively; and the proportion of adsorbed water in coal pores and fractures decreases to 76.808%. These data mark a notable reduction in water wettability.
- 2) When the CO₂ temperature falls, its adsorption capacity becomes less sensitive to temperature changes, with only a 2.93 cm³/g difference observed between 353K and 393K. The moderate activation effect of the temperature field enhances the probability that water molecules interact with pore walls and get adsorbed onto their surfaces. At 373K, the coal-water adhesion work reaches a peak of 58.01 mN/m; the surface energy of coal rises to 39.48 mN/m; and the proportion of adsorbed water in pores and fractures grows to 78.438%. These results reflect an improvement in water wettability.
- 3) When the temperature of CO₂ is further reduced, the thermal activation effect on water molecules caused by the gas–water temperature difference becomes negligible. At 333.00K, the slope of the MSD of water molecules is only 0.00185, indicating a low level of molecular activity. Meanwhile, the CO₂ adsorption capacity increases significantly at this temperature, reaching 35.31 cm³/g. Under the influence of competitive wetting by CO₂, adsorbed water molecules in the coal matrix are displaced from their adsorption sites

and migrate outward pore-fracture space. As a result, the proportion of adsorbed water in the pore-fracture space decreases from the initial 98.968%–78.052%, indicating a deterioration in water-wettability.

By comparing the analysis results of water wettability changes obtained using two different methods, namely contact angle measurement and coal-water interface characteristic parameters, a conclusion is drawn that the three-phase contact angle θ cannot serve as a sole basis for accurately judging changes in water wettability within the micropores and fractures of coal with the rise of CO₂ injection temperature. Despite its wide use by many scholars to assess the water wettability of coal, the contact angle measurement method has certain limitations. In the wetting system studied herein, the contact angle can only effectively reflect changes in droplet wettability caused by competitive adsorption of CO₂ molecules on the coal surface.

CRediT authorship contribution statement

Shunqing Ma: Writing – review & editing, Writing – original draft, Software, Resources, Data curation, Conceptualization. **Baiquan Lin:** Writing – review & editing, Methodology, Funding acquisition. **Jiajia Zhao:** Writing – review & editing, Supervision. **Xiangliang Zhang:** Writing – review & editing, Project administration. **Qian Liu:** Software, Resources. **Ting Liu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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